LATENT STATE SPECTRAL TRANSFER: EIGENSPACE ALIGNMENT BEYOND EIGENVALUES

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ABSTRACT

Spectral analyses of learned dynamics commonly use the eigenvalues of a latent predictor to characterize which modes of a representation are preserved, attenuated, or amplified over time. This interpretation is incomplete, however, because eigenvalues determine an operator only up to a change of basis and therefore do not specify how its modes are oriented relative to the covariance structure of the latent states on which it acts. We formulate latent transfer as a property of the operator pair (Σ, P), where Σ is the latent covariance and P is the predictive transition. To isolate the missing geometric dependence, we construct an isospectral family $P_\theta = R_\theta \Lambda R_\theta^\top$ whose eigenvalues remain fixed while its eigenspaces rotate relative to those of Σ. For a rotation restricted to coordinates (i, j), we prove the exact identity $\|[\Sigma, P_\theta]\|_F = |(\sigma_i - \sigma_j)(\lambda_i - \lambda_j)\sin(2\theta)|/\sqrt{2}$. The result establishes that a single predictor spectrum is compatible with a continuum of covariance-relative transfer geometries: misalignment is maximal at an oblique orientation, vanishes for jointly diagonal configurations, and disappears when either operator is locally isotropic. Controlled analyses confirm the predicted angular dependence and show that eigenspace rotation at fixed eigenvalues changes covariance-weighted energy transport and the accumulation of multi-step prediction residuals. Comparisons based on spectrum-preserving permutations, randomized bases, eigengap interventions, dimensional variation, and alternative transfer measurements distinguish this effect from spectral shape, coordinate ordering, and metric choice. We further evaluate the same geometric quantities in representations produced by LeWorldModel, I-JEPA, JEPA-WM, and DINO-WM. Across image transformations and action-conditioned temporal transitions, the observed covariance-relative structure is weakened when correspondence is disrupted while representation scale or operator spectrum is preserved, indicating that the measurements capture relational organization not recoverable from eigenvalues alone. These findings yield a general methodological conclusion: spectral accounts of latent transfer require either an explicit shared-basis assumption or a direct measurement of covariance–predictor alignment, such as a commutator norm, principal angle, or spectral-projector overlap. The proposed diagnostic assumes no predefined transformation group and is therefore symmetry-agnostic and complementary to equivariant predictive representation learning.

1 INTRODUCTION

Latent predictive models compress observations into representations on which future states, masked regions, or transformed views can be predicted. Their high-dimensional behavior is often summarized spectrally: covariance eigenvalues describe where representation energy is concentrated, while predictor eigenvalues describe the persistence or attenuation of transition modes. Spectral descriptions are central to analyses of frequency bias and operator learning (Rahaman et al., 2019; Cao et al., 2019; Tancik et al., 2020; Hong et al., 2022; Li et al., 2021; Khodakarami et al., 2026). Yet a latent predictor acts on a representation distribution, not in isolation. The scientific question is therefore relational: do its eigenvalues identify which covariance-weighted latent directions are actually transported?

Eigenvalues determine an operator only up to its eigenbasis and contain no information about its orientation relative to latent covariance. Two predictors can consequently share an identical eigenvalue

list while transporting high-variance directions differently. Mapping predictor eigenvalues directly to transfer thus requires an additional condition—a known shared basis, joint diagonalizability, or measured near-commutativity. We formalize this dependence through the pair (Σ_z, P), where Σ_z is latent covariance and P is a predictive transition or view-induced action. Commutators, principal angles, and spectral-projector overlaps measure the relation between their invariant subspaces. Figure 1 illustrates the resulting question in an action-conditioned LeWorldModel transition.

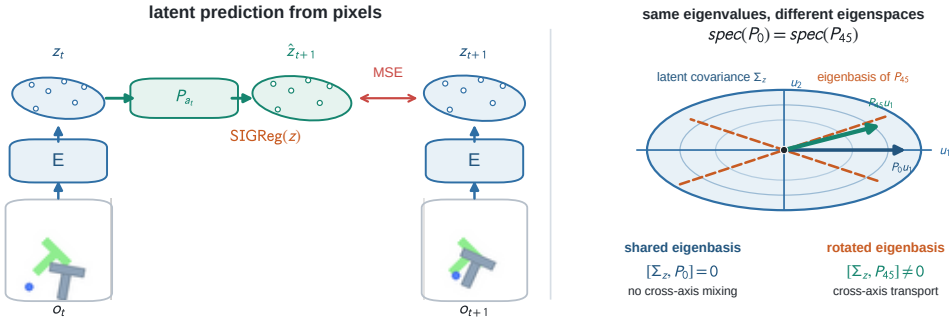


Figure 1: One learned-system instance of the operator pair. PushT simulator frames are encoded into adjacent LeWM latents; the action-conditioned predictor produces $\hat{z}_{t+1}$ under prediction MSE and SIGReg. The resulting latent covariance Σ_z and local transition action P instantiate the same operator-pair question studied in the theorem. The same transition spectrum can be aligned or rotated relative to Σ_z, yielding commuting or noncommuting covariance-relative transport. The geometry is schematic; quantitative changes appear in Figure 2.

Our theoretical result fixes diagonal spectra Σ and Λ and rotates only the predictor eigenspace, defining $P_\theta = R_\theta \Lambda R_\theta^\top$. Every P_θ is isospectral, yet we prove that $\|[\Sigma, P_\theta]\|_F$ is proportional to $|(\sigma_i - \sigma_j)(\lambda_i - \lambda_j)\sin(2\theta)|$. The mismatch vanishes at jointly diagonal endpoints, peaks at $45°$, and disappears when either spectrum is flat on the rotated plane. A spectrum-only account predicts invariance across this family; an alignment account predicts the closed-form oblique response and corresponding variation in covariance-weighted transfer. The construction therefore gives both a non-identifiability theorem and a falsifiable signature of the missing geometry.

The experiments implement this discriminator. The primary sweep varies only relative eigenspace orientation while holding predictor eigenvalues fixed. Jointly diagonal permutations, spectrum-matched random bases, and flat-spectrum cases test coordinate ordering, arbitrary noncommutativity, and eigengap dependence. Ablations over anisotropy, dimension, filtering, and readout test numerical robustness. The commutator follows the predicted $|\sin(2\theta)|$ law, while covariance-weighted energy transfer and multi-horizon residual growth vary under the same fixed-spectrum perturbation. These controls isolate alignment from spectral shape and metric choice.

We next evaluate the same quantities in learned representations. Frozen LeWorldModel latents and I-JEPA features exhibit covariance-relative transition and view-displacement structure beyond randomized controls. Released JEPA-WM and DINO-WM checkpoints provide explicit action-conditioned operators, evaluated through cross-fitting, pair-shuffled and spectrum-preserving controls, reconstructed splits, and action-order interventions. The theorem establishes finite-dimensional non-identifiability; these measurements show that alignment, temporal correspondence, prediction error, and action dependence remain empirically separable.

The contributions of this work are threefold:

- We formulate spectral transfer as an identification problem for (Σ_z, P) and prove that predictor eigenvalues alone are insufficient.
- We derive a closed-form commutator law and controls that separate alignment from spectral and basis confounds.
- We validate the diagnostics in frozen representations and action-conditioned world models using randomized and held-out controls.

The resulting contribution is a measurement principle: a predictor spectrum identifies latent transfer only with a justified shared-basis condition or an explicit covariance–predictor alignment statistic. This principle is symmetry-agnostic and complementary to architectures that impose invariance or equivariance. It determines when a spectral explanation is sufficient and what additional relation must be reported when it is not.

2 Related Work

Spectral bias and learned operators. Spectral accounts of learning describe mode-dependent approximation, convergence, and generalization. Neural networks often fit low-frequency components before high-frequency components (Rahaman et al., 2019; Cao et al., 2019), while kernels, Fourier features, architectures, and activations alter the resulting frequency response (Jacot et al., 2018; Tancik et al., 2020; Hong et al., 2022; Fridovich-Keil et al., 2022). The same perspective is central to neural operators: Fourier Neural Operators parameterize mappings in spectral coordinates (Li et al., 2021), and recent analyses study the loss, recovery, or rescaling of high-frequency content (Khodakarami et al., 2026; Perrone et al., 2026; Xie & Luo, 2026; Sun et al., 2026). These works ask how behavior varies across spectral modes. We ask what remains unidentified when those eigenvalues are held fixed.

Latent predictive representations. World models learn compact dynamics for prediction, planning, and control (Ha & Schmidhuber, 2018; Hafner et al., 2020; 2023; Micheli et al., 2023; Hansen et al., 2024), whereas image and video JEPAs learn predictive representations without reconstructing pixels (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025). Recent work has extended these ideas to latent planning, temporally extended prediction, stable end-to-end training, and joint-embedding predictive world models (Maes et al., 2026; Zhang et al., 2026; Du et al., 2026; Ding et al., 2026; Terver et al., 2025). In all of these systems, a latent covariance describes where representation energy lies and a predictor describes how that representation changes. Our concern is the geometry between those two objects, rather than either spectrum considered separately.

Equivariant predictive world models. The closest JEPA precedent to our geometric question is seq-JEPA, which processes short sequences of transformed views, conditions prediction on relative transformations or actions, and uses architectural inductive biases to separate an equivariant view representation from an invariant aggregate representation (Ghaemi et al., 2025). Our objective is different: we do not propose an equivariant architecture, assume a transformation group, or use the commutator as an equivariance certificate. We introduce a *symmetry-agnostic diagnostic* for latent predictive systems: whether a predictor's eigenvalue list is sufficient to identify its transfer relative to latent covariance. Thus our analysis is *complementary to equivariant JEPA*: equivariant JEPA specifies or induces how representations transform under known actions, whereas our covariance–predictor alignment test asks what must be measured about the resulting operator pair even when symmetry is absent, partial, or unspecified.

Representation alignment. Several lines of work already separate spectral shape from basis alignment. Spectral prediction theory decomposes representation prediction error into eigenspectrum and alignment terms (Canatar et al., 2023); latent functional maps compare representation spaces through spectral geometry (Fumero et al., 2024); and cross-modal diagnostics can reveal similar eigenvalue spectra despite poorly aligned eigenvector bases (Sarkar, 2026). Subspace-alignment methods likewise compare learned spaces through basis-sensitive quantities (Fernando et al., 2014; Thopalli et al., 2022). We specialize this distinction to the covariance–predictor pair and connect it directly to transfer readouts under an exactly fixed predictor spectrum.

Commutators and invariant subspaces. Matrix analysis provides the appropriate language for this pairwise question. Principal angles, spectral projectors, and perturbation bounds describe relations between invariant subspaces (Bhatia, 1997; Stewart & Sun, 1990; Björck & Golub, 1973; Davis & Kahan, 1970; Knyazev & Argentati, 2002). For normal matrices, commutativity is equivalent to joint diagonalizability; a nonzero commutator therefore witnesses the absence of a shared eigenbasis, a distinction also used in statistical tests of simultaneous diagonalization (Xu et al., 2024) and noncommutative filtering (Parada-Mayorga & Ribeiro, 2021). We turn this standard algebraic

distinction into a spectral-transfer identification test by constructing an isospectral predictor family and deriving its commutator curve in closed form.

3 METHOD

Finite-dimensional transfer model. We study finite-dimensional latent transfer through a covariance operator $\Sigma \in \mathbb{R}^{d\times d}$ and a predictor operator $P \in \mathbb{R}^{d\times d}$. The covariance assigns energy to latent directions, while the predictor transports that energy through a one-step transition. The spectrum of P identifies transfer only if all predictors with that spectrum induce the same covariance-relative transfer functional.

Same-spectrum construction. Let

$$\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_d), \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d). \tag{1}$$

For coordinates $i \neq j$, let R_θ be the identity outside the (i, j) plane and a two-dimensional rotation inside that plane. The same-spectrum predictor family is

$$P_\theta = R_\theta \Lambda R_\theta^\top. \tag{2}$$

All members of the family in Eq. (2) have spectrum $\{\lambda_1, \ldots, \lambda_d\}$. The construction therefore changes predictor eigenspaces while holding predictor eigenvalues fixed. Any quantity that changes with θ is not determined by the predictor spectrum alone.

Eigenspace-alignment measurements. We measure covariance-predictor eigenspace mismatch by

$$\mathcal{C}(\theta) = \frac{\|[\Sigma, P_\theta]\|_F}{\|\Sigma\|_F \|P_\theta\|_F}. \tag{3}$$

The normalized commutator in Eq. (3) vanishes when the two operators commute and increases with their covariance-relative eigenspace mismatch.

The same relation can be described through spectral projectors. If Π_a^Σ and Π_b^P denote covariance and predictor eigenspace projectors, then

$$\mathcal{O}_{ab} = \|\Pi_a^\Sigma \Pi_b^P\|_F^2 \tag{4}$$

records eigenspace overlap, equivalently the squared cosines of principal angles for simple subspaces. We use Eq. (3) in the main theorem because it admits a closed-form curve; Eq. (4) provides the corresponding subspace-level vocabulary for learned representations.

Transfer readouts. We use two transfer readouts. For a top coordinate band B_k, the top-band lift compares predicted energy in B_k to the isotropic baseline,

$$\mathcal{T}_{k,h}(P_\theta) = \frac{\mathrm{tr}(\Pi_k P_\theta^h \Sigma (P_\theta^h)^\top)}{\mathrm{tr}(P_\theta^h \Sigma (P_\theta^h)^\top)} \Big/ \frac{k}{d}, \tag{5}$$

where Π_k projects onto the first k coordinates. Residual growth compares residual trace at a long horizon to residual trace at the first horizon under the same synthetic generator. These deterministic readouts quantify covariance-relative transfer across horizons.

The diagonal-transfer generator used by the analytical sweeps gives this residual readout a concrete form. Let ρ_j be the coordinate-j predictor response and let $\tilde{\sigma}_j$ be the heat-filtered latent variance. At horizon h, the predicted trace and residual trace are

$$\mathcal{P}_h = \sum_{j=1}^{d} \tilde{\sigma}_j \rho_j^h, \tag{6a}$$

$$\mathcal{E}_h = \sum_{j=1}^{d} \tilde{\sigma}_j (1 - \rho_j^h)^2. \tag{6b}$$

We summarize multi-horizon accumulation in Eq. (6b) by $\mathcal{E}_8/\mathcal{E}_1$, the ratio of long-horizon to one-step residual trace.

4 THEORETICAL ANALYSIS

Theorem 1 (Single-Plane Same-Spectrum Commutator Identity)**.** *Let Σ and Λ be diagonal. Let $P_\theta = R_\theta \Lambda R_\theta^\top$, where R_θ rotates only coordinates (i, j). Then*

$$\|[\Sigma, P_\theta]\|_F = \frac{|(\sigma_i - \sigma_j)(\lambda_i - \lambda_j)\sin(2\theta)|}{\sqrt{2}}. \tag{7}$$

After normalization by $\|\Sigma\|_F \|P_\theta\|_F$, the same $|\sin(2\theta)|$ dependence remains. The curve is zero at $\theta = 0°$ and $\theta = 90°$, maximized at $\theta = 45°$, and identically zero if either spectrum is flat on the rotated plane.

Theorem 1, through the identity in Eq. (7), gives a direct identification result. The eigenvalues of P_θ are invariant in θ, so any statistic that changes with θ cannot be determined by the predictor spectrum alone. The commutator is the lowest-order witness that the covariance and predictor are not jointly diagonal in the same basis.

Proof sketch. Only the rotated two-dimensional block contributes to the commutator. Write $c = \cos\theta$, $s = \sin\theta$, $\Delta_\sigma = \sigma_i - \sigma_j$, and $\Delta_\lambda = \lambda_i - \lambda_j$. In the rotated plane,

$$P_{\theta,ij} = \begin{bmatrix} \lambda_i c^2 + \lambda_j s^2 & \Delta_\lambda cs \\ \Delta_\lambda cs & \lambda_i s^2 + \lambda_j c^2 \end{bmatrix}, \tag{8}$$

while Σ_{ij} remains diagonal. Substituting Eq. (8) into the commutator gives

$$[\Sigma, P_\theta]_{ij} = \begin{bmatrix} 0 & \Delta_\sigma \Delta_\lambda cs \\ -\Delta_\sigma \Delta_\lambda cs & 0 \end{bmatrix}. \tag{9}$$

Taking the Frobenius norm of Eq. (9) gives $2(\Delta_\sigma \Delta_\lambda cs)^2 = (\Delta_\sigma \Delta_\lambda)^2 \sin^2(2\theta)/2$. All other blocks vanish, and orthogonal conjugation keeps $\|P_\theta\|_F = \|\Lambda\|_F$, so normalization does not change the angular dependence.

Proposition 1 (Same-Spectrum Ambiguity)**.** *Assume $(\sigma_i - \sigma_j)(\lambda_i - \lambda_j) \neq 0$. The family $\{P_\theta : \theta \in [0, \pi/2]\}$ has a constant predictor spectrum but a nonconstant alignment statistic $\mathcal{C}(\theta)$. Therefore the predictor eigenvalue list alone cannot identify commutator-sensitive covariance-predictor alignment for the pair (Σ, P_θ).*

Proposition 2 (Fixed-Spectrum Transfer Readout Variation)**.** *In the same two-dimensional rotated plane, let Π_i project onto coordinate i. Define the horizon-one coordinate energy and its closed form by*

$$Q_i(\theta) = \mathrm{tr}(\Pi_i P_\theta \Sigma P_\theta^\top). \tag{10a}$$

$$\begin{aligned} Q_i(\theta) = {} & \sigma_i(\lambda_i \cos^2\theta + \lambda_j \sin^2\theta)^2 \\ & + \sigma_j(\lambda_i - \lambda_j)^2 \cos^2\theta \sin^2\theta. \end{aligned} \tag{10b}$$

Thus the coordinate-energy readout in Eq. (10b) *varies with θ while the predictor eigenvalues remain fixed.*

The identity also specifies when the effect disappears. If the latent covariance or predictor spectrum is flat on the rotated plane, the commutator is zero for every angle. A coordinate permutation of a jointly diagonal pair likewise preserves commutativity. A random basis can create a nonzero commutator, but need not reproduce the structured oblique peak of a single-plane rotation. These cases motivate the controls below.

Corollary 1 (What Spectra Can and Cannot Certify)**.** *Spectrum shape controls the scale of the alignment effect through the eigengaps $(\sigma_i - \sigma_j)$ and $(\lambda_i - \lambda_j)$, but the predictor spectrum alone cannot certify the effect. A complete spectral-transfer analysis must specify at least one pairwise relation between the latent geometry and transition operator, such as a commutator norm or eigenspace-overlap statistic.*

Eigenvalue decay describes how mass is distributed over predictor modes; eigenspace alignment determines whether those modes coincide with the covariance directions whose transfer is being interpreted. The two quantities answer different questions and must be reported accordingly.

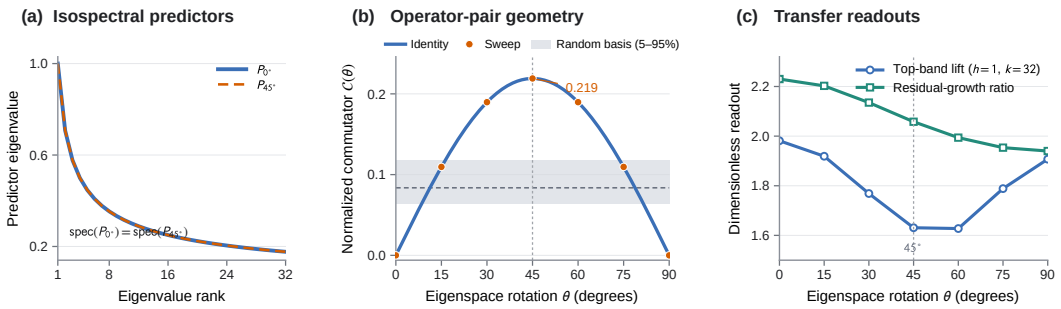


Figure 2: Same eigenvalues do not imply the same transfer. The left panel shows that the aligned predictor P_{0° and oblique predictor P_{45° have the same eigenvalue list. The center panel rotates only the predictor eigenspace relative to the covariance eigenspace, raising the normalized commutator from 0.000 to 0.219 and tracing the exact $|\sin(2\theta)|$ witness above the spectrum-matched random-basis null. The right panel shows the resulting changes in top-band lift and residual growth, motivating alignment statistics alongside eigenvalue reports.

5 Experiments

The analytical experiments are deterministic, allowing the spectrum-only explanation to be tested without estimation noise. We use $d = 64$, power-law latent and predictor spectra, rotation angles $\{0, 15, 30, 45, 60, 75, 90\}$ degrees, horizons $\{1, 2, 4, 8\}$, and top bands $k \in \{4, 8, 16, 32\}$. The rotated plane is formed by the first and last coordinates, with latent and predictor eigengaps 0.984375 and 0.875. Within each sweep, the quantities not under study are fixed by construction.

The comparison design separates basis alignment from spectral shape. The angle sweep varies only the predictor basis; the permutation-preserving and random-basis controls retain the predictor spectrum while changing the way it is positioned; and the factor-isolating ablations hold either the spectrum or the alignment angle fixed. Additional readouts, dimensions, and nuisance perturbations are reported in the supplement.

The learned-representation experiments instantiate the operator-pair question at two levels. First, frozen LeWorldModel PushT latents and frozen I-JEPA image features test covariance-relative displacement measurements without fitting a temporal operator. The PushT analysis compares dynamics, planner-weighted dynamics, prediction residuals, and transition overlap with norm-matched controls, label shuffles, and a random-subspace null. The I-JEPA analysis measures whether deterministic view displacements concentrate in the covariance eigenspace of source features across six vision benchmarks.

Second, released JEPA-WM and DINO-WM checkpoints test explicit action-conditioned operator estimates. The four checkpoint–environment combinations comprise JEPA-WM and DINO-WM on PushT and PointMaze at horizons $h \in \{1, 2, 4\}$. Full-rank commutators are paired with cross-fitted low-rank operators, pair-shuffled and spectrum-preserving controls, within-episode action-order derangements, and alternative PointMaze memberships. Together these comparisons separate prediction error, temporal correspondence, and covariance–operator alignment.

6 Results and Analysis

We first test identifiability with the isospectral construction, then isolate eigenspace alignment with controlled perturbations, and finally examine how the same measurements behave in frozen representations and held-out temporal operators. This sequence moves from an exact algebraic statement to its empirical measurement consequences.

Figure 2 exhibits the identification failure under a fixed spectrum. The commutator rises from zero at 0° to 0.219063 at 45° and returns to numerical zero at 90°, although the predictor eigenvalues never change. The structured peak also exceeds the random-basis reference, whose mean is 0.0836631 and whose 5th–95th percentile interval is $[0.0637529, 0.11723]$. Transfer changes with the same rotation: for top-$k = 32$, horizon-1 top-band lift falls from 1.98127 at alignment to 1.63077 at 45°,

Table 1: Spectrum-preserving basis perturbations. All rows keep the same predictor eigenvalues. Larger $\mathcal{C}$ indicates greater covariance–predictor eigenspace mismatch. The percentile columns summarize repeated random-basis draws; deterministic constructions have degenerate percentile intervals.

Basis construction	Mean $\mathcal{C}$	5th percentile	95th percentile
Oblique plane rotation	0.219063	0.219063	0.219063
Permutation-preserving null	0.000000	0.000000	0.000000
Random-basis null	0.084512	0.061633	0.121488

Table 2: Alignment and spectral anisotropy as distinct factors. The upper block fixes the predictor spectrum and varies only the basis angle. The lower block fixes the angle at 45° and varies input spectral anisotropy; larger stable rank denotes a flatter latent spectrum.

Isolation axis	Perturbation	Latent stable rank	Normalized commutator $\mathcal{C}$	Residual ratio $\mathcal{E}_8/\mathcal{E}_1$
Fixed spectrum	0° angle	4.74389	0.000000	2.23004
Fixed spectrum	45° angle	4.74389	0.219063	2.05798
Fixed spectrum	90° angle	4.74389	0.000000	1.93994
Fixed alignment	exponent 0.0	14.60210	0.114121	1.78340
Fixed alignment	exponent 1.0	4.74389	0.219063	2.05798
Fixed alignment	exponent 2.0	2.36335	0.258604	2.19337

while residual-growth ratio changes from 2.23004 to 2.05798. Predictors that are indistinguishable by eigenvalues are therefore distinguishable by covariance-relative transfer.

The full rotation table in the supplement follows the theorem's exact symmetric curve across all tested angles: 0.000000, 0.109531, 0.189714, 0.219063, 0.189714, 0.109531, and 0.000000 from 0° to 90°. The projector-based transfer readouts additionally inherit the ordered covariance coordinates, so they distinguish rotations on opposite sides of 45° even when the commutator magnitudes agree. This difference links the basis-invariant mismatch witness to the coordinate-specific transfer quantity of interest.

Table 1 isolates the alignment mechanism. A permutation-preserving construction keeps the operators jointly diagonal and collapses the commutator to zero, separating the signal from coordinate reordering. A spectrum-matched random basis creates nonzero noncommutativity, but its mean and 95th percentile remain below the structured oblique peak. The structured rotation therefore produces a distinct alignment response beyond spectrum matching or arbitrary basis randomization.

Table 2 separates alignment from spectral anisotropy. Under fixed spectrum, alignment alone moves the commutator range by 0.219063. Under fixed 45° alignment, changing the input exponent from 0 to 2 changes stable rank from 14.60210 to 2.36335 and peak commutator from 0.114121 to 0.258604. Spectra therefore control effect size, but they do not determine whether the pair of operators commute.

The same conclusion persists under alternate readouts and dimensions. The $0^\circ \to 45^\circ$ basis perturbation changes both the $h = 1$ top-band lift ($1.98127 \to 1.63077$) and the $h = 8$ residual-band lift ($1.62670 \to 1.55638$). Cross-dimension sweeps at $d = 64, 128, 256$ preserve the predictor-exponent ordering in residual growth, while the $h = 8/h = 1$ residual ratio for predictor exponent 0.5 changes from 4.56383 at $d = 64$ to 3.34861 at $d = 256$. Increasing the predictor exponent from 0.5 to 2.0 lowers this ratio from 3.85419 to 1.31785 in the $d = 128$ analytical generator. Heat-filter recovery remains stable under small noise, with mean known-generator R^2 changing from 1.000000 at noise standard deviation 0.00 to 0.999498 at 0.05.

The controls identify the theorem's operating conditions. Flat latent or predictor spectra remove the eigengap that drives the oblique signal; permutation-preserving bases retain joint diagonalization; and the metric ablations show that the transfer interpretation persists across readouts. The analytical results therefore separate eigenspace alignment from coordinate ordering, spectral flatness, and metric choice.

6.1 Frozen LeWorldModel PushT Latent Measurements

We next evaluate the same alignment-sensitive readouts on frozen LeWorldModel PushT latents. The question is whether learned latents exhibit covariance-relative structure that an eigenvalue summary would miss. The analysis is offline, uses stored latent records, and reports top-$k = 32$ readouts against randomized controls.

The frozen PushT measurements show a clear separation from randomized structure. Dynamics, planner-weighted, predicted-dynamics, and residual lifts are 4.71084, 4.62362, 4.68678, and 3.68668, compared with a norm-matched random value of 1.00082. Action, state-cluster, and time shuffles reduce the dynamics readout to 1.86731, 1.86817, and 2.18068. The mean transition overlap is 0.0299704, above the random-subspace mean of 0.0209526, and planner weights place $1.44465\times$ more mass in the top band than in the tail band. An independent three-seed, within-cell held-out check gives positive cell-level action-to-dynamics R^2 means from 0.05502 to 0.06034, whereas action-row permutation nulls are negative (from -0.03469 to -0.02422). This supports detectability in the frozen representation under the stated readout, not causal controllability or online planning; the full protocol and cell ranges are reported in the supplement.

6.2 Frozen I-JEPA Feature Measurements

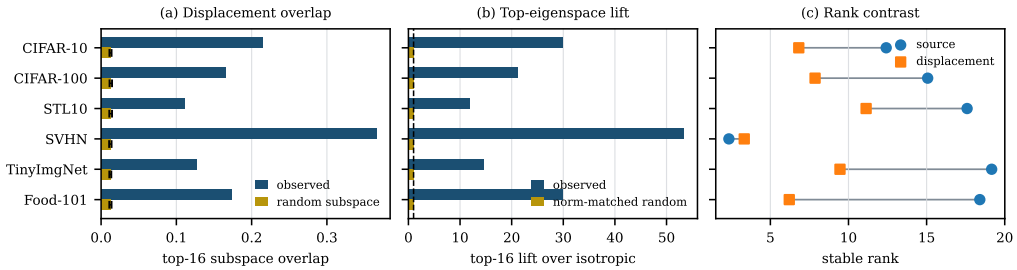


Figure 3: Frozen I-JEPA features show covariance-relative view-transfer structure across six standard vision benchmarks. Thirty balanced images per benchmark are encoded with the official frozen I-JEPA ViT-H/16 model under horizontal flips. Panels compare source/displacement top-16 overlap against a random-subspace null, displacement top-eigenspace lift against a norm-matched random-direction control, and source versus displacement stable rank.

This frozen I-JEPA experiment evaluates covariance-relative view transfer across six standard vision benchmarks. The top-16 source/displacement overlap ranges from 0.11062 to 0.36572, compared with random-subspace means near 0.0124. The corresponding top-16 energy lifts range from 12.01 to 53.33, while norm-matched random directions remain near one. The effect is therefore shared across datasets but not constant in magnitude, consistent with a covariance-relative rather than spectrum-only description of the view displacement.

An operator-fit stress test complements this positive view-transfer result. Across five held-out fits (CIFAR-10 with ranks 8 and 16; scikit-learn-digits with ranks 8, 16, and 32), held-out R^2 ranges from 0.97141 to 0.99840, while the fitted commutator stays below the spectrum-preserving random-basis null mean in every fit (differences from -0.00417 to -0.09323). View predictability and covariance-relative operator identification therefore separate as estimands even within frozen JEPA features.

6.3 Held-out Action-Conditioned World Models

Full-rank commutators reveal strong regime dependence but do not by themselves establish out-of-sample temporal correspondence. On PushT, both JEPA-WM and DINO-WM move from below their spectrum-preserving references at short horizons to above them at $h = 4$; on PointMaze, both models remain above the reference across $h = 1, 2, 4$. We therefore fit rank-16 ridge operators on training folds and evaluate held-out episodes. Paired R^2 exceeds pair-shuffled R^2 at every PointMaze horizon: $0.082/0.183/0.025$ versus $-0.092/-0.094/-0.081$ for JEPA-WM, and $0.935/0.808/0.554$ versus

$-0.093/-0.083/-0.065$ for DINO-WM. Prediction error, commutator magnitude, and held-out temporal correspondence are thus distinct properties of the learned transition.

Action-order interventions expose where this correspondence uses the supplied future actions. Deranging five-step action blocks within each episode increases PushT prediction error for JEPA-WM at all horizons and for DINO-WM once a deranged future block enters the rollout. DINO-WM PointMaze remains sensitive at horizons 2 and 4, whereas JEPA-WM PointMaze does not deteriorate under the same intervention. Action-order sensitivity is therefore a property of a particular checkpoint, environment, and horizon, rather than an automatic consequence of a nonzero commutator.

The held-out correspondence is stable to the reconstructed PointMaze membership. Across three splitter seeds and a disjoint tail holdout, the paired-minus-shuffled R^2 gap remains positive for both models at every horizon. At $h = 8$, however, the JEPA-WM PointMaze paired score is -0.040 versus -0.044 after shuffling, leaving no clear positive out-of-sample signal. The temporal evidence therefore supports a scoped measurement conclusion: alignment and correspondence are measurable beyond one split, but their strength and action dependence vary with the estimator and rollout regime.

7 Discussion and Limitations

The central consequence is an operator-pair measurement principle. A predictor spectrum describes amplification or suppression in the predictor's own eigenbasis, whereas a latent covariance spectrum describes variation in the representation. The commutator, principal angles, and spectral-projector overlaps quantify the relation between these two geometries. Eigenvalue-only reporting becomes sufficient under an additional condition such as simultaneous diagonalization, a known shared eigenbasis, or a measured small commutator.

This perspective is *complementary to equivariant JEPA* designs such as seq-JEPA (Ghaemi et al., 2025). Such models use known relative transformations and sequence architecture to induce separate equivariant and invariant representations. Our diagnostic is *symmetry-agnostic*: it neither assumes nor tests a particular group representation. An equivariance law can constrain how a transformation acts without, by itself, making the predictor spectrum sufficient to characterize its relation to the latent covariance. The two analyses therefore answer different questions and can be used together: equivariance constrains transformation structure, while commutator and projector measurements assess whether the predictor spectrum is sufficient to describe transfer in the learned latent geometry.

The learned-representation measurements instantiate this principle at three scales. Frozen LeWM PushT latents exhibit top-band and transition-overlap structure relative to randomized controls. Frozen I-JEPA features exhibit covariance-relative view-displacement structure across six benchmarks, while held-out operator fits separate view predictability from stable operator identification. Released JEPA-WM and DINO-WM checkpoints further separate same-sample commutator scale, held-out temporal correspondence, and sensitivity to action order.

These results establish the portability of the estimands, rather than a universal ordering for learned models. The learned operators are local linear summaries of a particular checkpoint, split, context window, and horizon; the commutator is a diagnostic of covariance-relative mismatch, not by itself a causal measure of planning or control quality. Establishing such a causal link requires matched interventions or downstream evaluations that vary alignment while controlling prediction accuracy. Within this scope, the operator-pair formulation converts a spectrum-only explanation into a testable hypothesis with an explicit estimand and null comparison.

The stronger external conditional-operator audit sharpens this boundary. Across 21 held-out PushT episodes, action-aligned magnitude and direction partitions produce greater local transition-map heterogeneity than a state-matched within-stratum permutation null under public, validation-split, and raw-action normalizations. The corresponding held-out R^2 contrast is not uniformly positive. We therefore interpret the result as evidence for action-conditioned latent regime structure beyond a coarse state-only explanation, not as a prediction gain, causal action effect, or planning improvement.

As a granularity check, the public-normalization result remains positive when the state-matched null uses 4, 8, or 16 first-coordinate strata: all four source–condition combinations retain positive 95% episode-bootstrap lower bounds. This supports robustness to the declared coarse-null resolution, not an identification of a unique physical state partition.

A matched intervention on frozen LeWorldModel latents, reported in the supplementary intervention section, makes this boundary explicit: conditional alignment changes the covariance–predictor diagnostic, but its paired one-step prediction effect remains negligible at small weight and reverses at a stronger weight. The result supports treating alignment as a measurable intervention target rather than as a universal proxy for predictive quality.

The external multi-step adapter and lag-control audits sharpen the same point: geometry interventions can be detectable while their rollout effect is non-monotone and normalization-sensitive, and the stronger future-versus-past directionality criterion is not uniformly supported across bands and horizons.

An independent eight-window replication of the frozen action-algebra audit provides a sharper temporal check. Across 21 held-out episodes and 164 windows, the action-permutation penalty is $+0.0254$ $[+0.0144, +0.0400]$ for encoded targets and $+0.0242$ $[+0.0148, +0.0359]$ for autoregressive predictions, whereas the corresponding one-step action gains, $+0.0082$ $[-0.0048, +0.0214]$ and $+0.0083$ $[-0.0034, +0.0202]$, include zero. The composition defect grows from 0.285 to 0.608 to 0.883 at $H = 2, 4, 8$ for encoded targets, with the same ordering for predictions. The composed-minus-direct MSE contrast is compatible with zero at $H = 2$ and $H = 4$ but is negative at $H = 8$. Thus the replication supports action-order sensitivity under a richer within-episode protocol, while the non-monotone direct-versus-composed result rules out interpreting it as an exact transition semigroup or an online planning result.

We added a leakage-controlled forward readout to distinguish this bounded diagnostic from a target-projection artifact. Using the released external PushT validation episodes, we predict a fixed source-defined dynamics target from only the current latent coordinates and action summaries, leaving one complete episode out of every fit. At $h = 8$, the dynamics top-8 input achieves $R^2 = 0.485$ with a state-plus-action gain of $+0.469$ (95interval $[+0.401, +0.536]$); its paired R^2 advantage over covariance, residual, and random top-8 inputs is $+0.562$, $+0.525$, and $+0.572$, respectively, with all intervals excluding zero. The result persists across three action normalizations and ridge values from 10^{-3} to 10^{-1}. This is the strongest current evidence that the extracted dynamics subspace contains forward-predictive information for an encoded latent band.

The horizon profile makes the result more specific. For the fixed dynamics target, the state-plus-action gain is $+0.056$ $[-0.013, +0.120]$ at $h = 1$, $+0.462$ $[+0.398, +0.536]$ at $h = 4$, and $+0.469$ $[+0.402, +0.532]$ at $h = 8$. Thus the evidence is not a one-step action correlation: the current dynamics coordinates remain predictive at the longer horizons, while replacing future actions with past actions lowers the joint readout at $h = 8$ by $+0.029$ $[+0.002, +0.062]$. The matched-band analysis also shows that the signal is distributed beyond the first eight coordinates: the dynamics middle band has gain $+0.250$ $[+0.170, +0.324]$ at $h = 4$ and $+0.431$ $[+0.385, +0.478]$ at $h = 8$. We therefore use top-8 as a compact, source-defined diagnostic rather than as a claim that all dynamics information is concentrated in the covariance-leading modes.

The same audit also defines the necessary negative boundary. When the target is the complete 192-dimensional latent increment rather than the fixed dynamics band, the dynamics top-8 state-plus-action gain is -0.038 ($[-0.059, -0.020]$). Thus the evidence supports a source-defined, band-level frozen predictive diagnostic, not universal full-latent transition prediction. A separate physical-displacement target gives an incremental R^2 of -0.120 ($[-0.174, -0.074]$) over a compact action summary for the dynamics top-8 predicted increment at $h = 8$. The encoded-latent result should therefore not be relabeled as additional physical-motion prediction; it does not establish physical controllability, causal action effects, online planning, or task success.

A rank-and-target-specificity check makes this separation more explicit. In the same leave-one-episode-out protocol, the dynamics-basis top-8 target receives an incremental R^2 of $+0.349$ $[+0.233, +0.461]$ at $h = 4$ and $+0.255$ $[+0.058, +0.461]$ at $h = 8$ from the current dynamics-basis coordinates beyond observed actions, whereas the covariance and random top-8 controls are negative at $h = 4$ (-0.453 $[-0.909, -0.188]$ and -0.278 $[-0.509, -0.068]$, respectively). The corresponding residual-basis target is selectively helped by residual-basis coordinates at $h = 8$ ($+0.279$ $[+0.124, +0.447]$), but not by dynamics-basis coordinates (-0.255 $[-0.445, -0.100]$). This target-specificity pattern supports separable encoded-latent subspaces; it does not imply that either subspace is a physical state or a causal control variable.

We then impose a stricter temporal transfer test: the readout is fitted at one horizon and evaluated at a different, held-out horizon, while the source bases remain fixed. For the predeclared forward transfer $h = 4 \rightarrow 8$, the dynamics top-8 input retains an incremental R^2 of $+0.364$ $[+0.156, +0.602]$ for the dynamics-band target, compared with -0.430 $[-0.765, -0.182]$ for the covariance basis and -0.188 $[-0.348, -0.039]$ for the random basis. The paired dynamics-minus-random contrast is $+0.553$ $[+0.351, +0.773]$, and the corresponding dynamics-minus-covariance contrast is $+0.794$ $[+0.441, +1.216]$. The residual-band target shows the complementary pattern: the residual top-8 input gives $+0.177$ $[+0.108, +0.252]$, whereas the dynamics input gives -0.112 $[-0.206, -0.025]$. Thus the stronger result is not merely a within-horizon fit: the source-defined dynamics and residual subspaces retain distinct predictive roles across a longer temporal scale. The reverse-horizon cells and the complete paired matrix are reported in the supplement because cross-horizon R^2 is not symmetric when the target scale changes. This remains a frozen latent readout, not a physical-dynamics, causal-control, online-planning, or task-success result.

We additionally stress-tested this transfer against action normalization and window density. Across the three normalization arms (public, validation-split z-score, and raw action divided by 100) and two, four, or eight windows per episode, the rank-8 dynamics-minus-random contrast remained positive in all three forward cells: the mean ranges were $+0.153$–$+0.185$ for $1 \rightarrow 4$, $+0.081$–$+0.121$ for $1 \rightarrow 8$, and $+0.386$–$+0.556$ for $4 \rightarrow 8$, with the corresponding episode-bootstrap lower bounds above zero. The $4 \rightarrow 8$ residual-minus-dynamics contrast likewise remained positive ($+0.218$–$+0.320$). This matrix supports robust cross-horizon predictive transfer under the declared preprocessing and sampling choices; it remains a frozen latent diagnostic rather than normalization-invariant causal dynamics, online planning, or task-success evidence.

As a source-fitting control, we repeated the transfer using two independently fitted dynamics/residual bases. Across both source splits, the three normalizations, and two or eight windows per episode, all 12 forward cells retained positive dynamics-minus-random lower bounds; the $4 \rightarrow 8$ residual-minus-dynamics contrast was also positive in every cell. The complete source-split audit is given in the supplement. This controls source-fit dependence while preserving the interpretation as a frozen latent readout.

Finally, we strengthened the observed-action control itself. The original readout uses a compact six-dimensional action summary; we repeated the audit with the complete sequence of per-block two-dimensional action means, padded to a common 16-dimensional representation for cross-horizon transfer. Across the six canonical cells and twelve source-fit cells, the richer-control dynamics-minus-random lower bound remained positive in every predeclared forward cell; the smallest lower bound was $+0.0278$, and the corresponding dynamics-minus-covariance lower bound was at least $+0.0499$. The $4 \rightarrow 8$ residual-minus-dynamics contrast also remained positive in all 18 cells. This addresses the possibility that the earlier result was created by an overly compressed action baseline. The richer history is still an observed-action predictive control, so the result supports latent predictive specificity beyond that control rather than a causal action effect. Details and the full matrix are given in the supplement.

We also ran a simulator-level closure that separates controllability, physical observability, and planning. A residual state-transition model fit on contact-covered simulator trajectories reaches held-out block-position errors of 2.36 and 3.54 at $h = 1$ for the two opposite-action arms, whereas the same model initialized from a frozen LeWM latent has errors of about 106 pixels. Errors grow to 30.12–30.18 for the state arm at $h = 8$, and neither the state nor visual model obtains task success in the matched four-episode MPC pilot. We then diagnosed this negative result rather than treating it as representation-level information absence. With 512 calibration endpoints and an XY-only Ridge readout, the observed-final latent block-position error falls to 19.062 pixels, but the predicted-final latent error remains 42.858 pixels. Removing single-frame-unidentifiable velocity from the visual state reduces the visual block-position error to $16.031/17.662$ for the two action arms at $h = 1$ and to $41.257/44.042$ at $h = 8$; the corresponding pose-only MPC still has zero success. The evidence therefore identifies a repairable calibration/state interface problem and a surviving contact-dynamics/planner problem. The full decoder, temporal-adapter, pose-only ablation, held-out counterfactuals, and planner controls are in the supplement. These results remain a closure diagnostic, not evidence of learned physical causality or accurate physical counterfactual prediction.

A candidate-ranking audit localizes the remaining planning failure further: the pose model's block-only ranking has Spearman correlation 0.063 with the simulator ranking, whereas a cost including

agent position and block angle reaches 0.974; nevertheless, the one-block random candidate set contains no reliably successful multi-stage push. The planner therefore needs both a contact-aware objective and a candidate protocol that represents approach followed by push. This is a design consequence of the measured closure gap, not evidence that the current frozen checkpoint already provides online planning.

We tested this diagnosis with a targeted compositional repair. The repair retained the frozen visual encoder and trained only the action-conditioned predictor modules on five contact-mode action-sequence arms per case, including piecewise and direction-changing sequences. On a disjoint 96/24 contact split, autoregressive $h = 8$ latent endpoint MSE decreased from 0.860 to 0.381. More importantly, a matched replay of 64 candidate sequences improved predicted-versus-actual task-cost Spearman correlation from 0.107 to 0.690 and reduced selected regret from 188.1 to 32.9. The oracle best candidate in that finite set still had task cost 169.6 (and 148.9 with 512 candidates), so the live contact-bootstrap MPC obtained zero task successes. This separates a repaired predictor-ranking interface from unresolved action coverage and multi-stage task control; it is not evidence that the original frozen checkpoint is causally physical or online-plannable.

We further separated first-contact protocol error from candidate support. A first-motion bootstrap can return while the agent is still carrying large velocity and has already passed the block. A settled-contact bootstrap with explicit braking, together with staged approach–push candidates, increased the best eight-step block progress from 49.217 to 60.231 pixels and reduced the best task cost from 271.865 to 264.421 on 16 held-out states. Nevertheless, the full-task success rate remained zero, as did the simulator-oracle success rate. At a 32-step open-loop horizon, block-pose success also remained zero on eight states. In live MPC, the privileged-state repaired run reduced endpoint pose error to 136.858 and final state distance to 176.457, while the matched visual-state run remained at 422.242 and 652.398, respectively; both had zero task successes. These controls show that the negative result is caused by a combination of contact bootstrap, state-interface, and multi-stage contact/orientation limitations, rather than by a single candidate-count or predictor-ranking defect. Full protocols and the claim boundary are given in the supplement.

We then repaired the simulator-state diagnostic itself. The audit identified three protocol failures that can otherwise masquerade as a representation failure: PushT's `n_contacts` counts wall and block collisions under the same collision type, the physical state contains agent velocity although a single pose omits it, and explicit state resets must be paired with the actual post-reset observation. We therefore defined contact by block-pose change, retained agent velocity in the adapter, matched reset geometry, and replaced hand-placed contact rings with genuine approach–contact trajectories. Across two independent seeds, the resulting simulator-state adapter reduced the $h = 8$ overall block-position error from 15.224 to 12.185 pixels in one seed and from 8.554 to 6.697 pixels in the other, relative to the constant-pose baseline. Contact-conditioned error nevertheless remained 38.991 and 24.727 pixels, and the two-seed closed-loop pilot achieved zero full-task and block-pose successes for every policy. The best block-aware two-block MPC reached 86.354 and 177.159 pixels of block XY distance in the two seeds, but this partial improvement did not transfer to task success. These results repair and sharpen the simulator-state diagnosis; they do not establish that the frozen LeWM latent supports accurate physical counterfactuals, online planning gain, or task success.

To separate image information from predictor information, we additionally fit matched physical readouts to the rendered image, the observed final latent, the frozen predictor's final latent, and the initial latent. The renderer-aware block-mask control reaches 40.396–47.599 pixels of $+a$ endpoint error across $h = 1, 2, 4$, whereas the observed latent reaches 43.728–50.985 and the predicted latent 48.394–63.822; at $h = 8$, the respective errors are 45.205, 57.832, and 58.780 pixels. The physical $+a/-a$ separation nevertheless grows from 2.167 to 45.484 pixels, while predicted latent separation remains between 9.914 and 12.890. This decomposition shows that physical position is visible in the rendered observation but is not yet a calibrated physical coordinate of the frozen latent predictor. The full protocol and uncertainty intervals are reported in the supplementary observability table; they do not upgrade the result to a learned physical causal model or an accurate physical counterfactual.

As a final repair test, we fit a downstream action-conditioned latent transition adapter using only the frozen encoder and contact-conditioned calibration trajectories. On two disjoint held-out splits, its $+a$ endpoint error was lower than the constant-pose baseline at $h = 2, 4, 8$ (for example, 36.7 versus 61.0 pixels and 29.7 versus 62.7 at $h = 2$), but remained above the observed-final-latent readout. The adapter therefore recovers part of the missing contact response without rescuing the original

predictor: the remaining failure is action-effect compression in the frozen predictor, not a mere decoder or contact-label bug. Full ablations and uncertainty intervals are reported in the supplement; this downstream repair is not retraining evidence or online-planning evidence.

We then tested the remaining predictor-side hypothesis with a separate contact-rich repair. Keeping the visual encoder and projector fixed, we supervised the action encoder, predictor, and prediction projection at every future latent step, while evaluating the repaired model autoregressively on held-out contact states. To avoid conflating predictor repair with a stale readout, the repaired latent distribution received its own calibration readout; all comparisons used the same paired $+a$ cases and two independent seeds. The calibrated $+a$ gain was $+32.8$ and $+37.0$ pixels at $h = 4$, and $+23.6$ and $+29.7$ pixels at $h = 8$ for seeds 8421 and 9271, respectively; the episode-bootstrap intervals are reported in the supplement and remain wide at $h = 8$. In contrast, applying the original readout to the repaired latent produced negative gains, showing that latent-distribution shift is a separate failure mode. The result supports a targeted repair of part of the action-effect compression, but it does not turn the original frozen checkpoint into a validated physical counterfactual model or establish online planning gain. The complete teacher-forced protocol, native-readout control, paired intervals, and seed-level rows are given in the supplement.

We further stress-tested this interpretation with a three-arm audit and an observed-latent upper bound. At $h = 4$, the calibrated gain was positive for all three matched arms ($+a$, $-a$, and zero) in both seeds; the $+a$ gains were $+32.8$ and $+37.0$ pixels, while the observed-latent upper-bound errors were 39.6 and 49.5 pixels. Thus the repair recovers part of the predictor/readout mismatch but leaves a substantial visual-observability ceiling. A calibration-size sweep also showed that eight contact cases are insufficient, whereas 32–48 cases approach the stable regime. Finally, a single-seed training-strength ablation gave h=4 gains of $+10.7$, $+32.8$, and $+53.7$ pixels after 1, 8, and 32 epochs, respectively, while 16 training cases produced a negative estimate. These latter ablations are reported as diagnostic sensitivity analyses rather than as additional multi-seed performance claims; together they explain why the unrepaired checkpoint cannot be judged from a single negative physical-readout result. An enlarged two-seed replication with 96 training and 24 held-out contact cases strengthens the $h = 4$ result: the calibrated $+a$ gains are $+37.7$ pixels (95% CI $[+19.8, +56.0]$) and $+31.0$ pixels (95% CI $[+18.8, +44.8]$). The corresponding $h = 8$ point gains remain positive but retain wider intervals, so we continue to treat long-horizon recovery as partial rather than complete. A matched 32-epoch, three-seed sensitivity audit further separates optimization from physical observability: all three $h = 4$ $+a$ intervals remain positive, whereas two of three $h = 8$ intervals still include zero. For seed 8421, latent endpoint MSE at $h = 8$ falls from 0.748 to 0.307 while the physical interval remains compatible with zero. Thus training longer improves latent prediction but does not by itself close the latent-to-physical contact-mode gap.

8 Conclusion

Predictor eigenvalues do not identify latent transfer without the eigenspaces on which they act. The finite-dimensional isospectral family proves this failure exactly, controlled perturbations isolate covariance–predictor alignment as the mechanism, and measurements in LeWM, I-JEPA, JEPA-WM, and DINO-WM show how the resulting estimands behave in learned representations. The practical rule is simple: a spectral-transfer analysis should state a shared-eigenbasis assumption or report an alignment statistic alongside the spectrum. This operator-pair view turns spectral description into a falsifiable account of which latent directions a learned transition actually transports.

Supplementary Material

The supplement includes the complete derivation and rotation sweep, spectrum-preserving nulls, factor-isolating and robustness analyses, frozen PushT and I-JEPA measurements, alignment-intervention boundaries, and full protocols for the held-out JEPA-WM and DINO-WM temporal evaluations.

Reproducibility Statement

The theory experiments are deterministic analytical simulations generated from the accompanying source code and stored numerical outputs. Frozen PushT measurements use stored offline latent records; I-JEPA measurements use the official frozen ViT-H/16 checkpoint and deterministic image-view pairs; temporal experiments use released JEPA-WM and DINO-WM checkpoints with the splits and preprocessing specified in the supplement. The accompanying package includes manuscript sources, figures, bibliography and style files, experiment scripts, numerical results, and compiled PDFs.

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